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THE VERY LOW ALBEDO OF WASP-12B FROM SPECTRAL ECLIPSE OBSERVATIONS WITH *HUBBLE*

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ABSTRACT

We present an optical eclipse observation of the hot Jupiter WASP-12b using the Space Telescope Imaging Spectrograph on board the Hubble Space Telescope. These spectra allow us to place an upper limit of $A_g < 0.064$ (97.5% confidence level) on the planet's white light geometric albedo across 290–570 nm. Using six wavelength bins across the same wavelength range also produces stringent limits on the geometric albedo for all bins. However, our uncertainties in eclipse depth are ∼40% greater than the Poisson limit and may be limited by the intrinsic variability of the Sun-like host star — the solar luminosity is known to vary at the 10^{-4} level on a timescale of minutes. We use our eclipse depth limits to test two previously suggested atmospheric models for this planet: Mie scattering from an aluminum-oxide haze or cloud-free Rayleigh scattering. Our stringent nondetection rules out both models and is consistent with thermal emission plus weak Rayleigh scattering from atomic hydrogen and helium. Our results are in stark contrast with those for the much cooler HD 189733b, the only other hot Jupiter with spectrally resolved reflected light observations; those data showed an increase in albedo with decreasing wavelength. The fact that the first two exoplanets with optical albedo spectra exhibit significant differences demonstrates the importance of spectrally resolved reflected light observations and highlights the great diversity among hot Jupiters.

Keywords: planets and satellites: atmospheres — stars: individual (WASP-12) — techniques: photometric

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基于*HUBBLE*光谱日食观测的WASP-12B极低反照率

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摘要

我们利用哈勃空间望远镜上的空间望远镜成像光谱仪对热木星WASP-12b进行了一次光学食观测。这些光谱数据使我们能够在290-570纳米波长范围内，对该行星的白光几何反照率设定上限为A_g <0.064（97.5%置信水平）。在同一波长范围内使用六个波长区间进行分析，也为所有区间得出了严格的几何反照率限制。然而，我们在食深度上的不确定性比泊松极限高出∼40%，这可能受到类太阳宿主星本身变化性的限制——已知太阳光度在分钟时间尺度上存在10^{-4}级别的变化。我们利用食深度限制检验了先前针对该行星提出的两种大气模型：氧化铝雾霾的米氏散射或无云瑞利散射。我们严格的未探测结果排除了这两种模型，并与热辐射加上来自原子氢和氦的弱瑞利散射相一致。我们的结果与另一颗温度低得多的热木星HD 189733b形成鲜明对比，后者是唯一一颗拥有光谱分辨反射光观测数据的热木星；那些数据显示反照率随波长减小而增加。首批两颗拥有光学反照率光谱的系外行星表现出显著差异，这证明了光谱分辨反射光观测的重要性，并凸显了热木星之间的巨大多样性。

Keywords: 行星与卫星：大气层 — 恒星：特定天体（WASP-12） — 技术：测光法

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1. INTRODUCTION

Thermal measurements of hot Jupiters suggest that these gas giant exoplanets often have moderate Bond albedos ($A_B \approx 0.4$, the fraction of incident energy reflected to space; Schwartz et al. 2017). However, many previous searches for reflected light from hot Jupiters have found little-to-none at optical wavelengths where the host star emits most of its energy (geometric albedo $A_g < 0.1$; e.g., Rowe et al. 2008; Kipping & Spiegel 2011; Heng & Demory 2013; Dai et al. 2017). It is unclear what is causing this apparent contradiction between constraints from thermal emission and optical reflection. Previous Hubble Space Telescope (*HST*) eclipse observations of HD 189733b with the Space Telescope Imaging Spectrograph (STIS) showed an increase in reflectivity toward bluer wavelengths which may, at least in part, explain the discrepancies between these two techniques (Evans et al. 2013).

A direct way to probe the back scattering efficiency of a hot Jupiter's atmosphere is observing the planet at optical wavelengths (where thermal emission is negligible) during eclipse, when the planet is near full phase and passes behind its host star. This method requires at least an order of magnitude higher photometric precision than transit observations of the same planet because the planet will be fainter than its host star, while the occulted area remains the same.

Observing an atmosphere at different orbital phases can provide further information about the scattering particles (e.g., Demory et al. 2013; Esteves et al. 2013; Heng & Demory 2013; Garcia Munoz & Isaak 2015; Shporer & Hu 2015; Oreshenko et al. 2016). Parmentier et al. (2016) suggested a connection between reflected light phase curve measurements and a sequence of condensate cloud models, but this only covered temperatures up to $T_{\rm eq} \sim 2200$ K: well below the equilibrium temperature of WASP-12b ($T_{\rm eq} = 2580$ K; Collins et al. 2017).

WASP-12b orbits a G0V star with an orbital period of 1.09 days (Hebb et al. 2009). While the host star is fairly faint (V = 12), WASP-12b's close semi-major axis and large radius ($a = 0.0234$ au, $R_p = 1.90\ R_J$, $R_p = 0.19\ R_*$; Collins et al. 2017) make it an excellent target for detailed study. Transit observations of WASP-12b range from 0.3 to 4.5 μm, and eclipse observations range from 0.9 to 8.0 μm (e.g., Hebb et al. 2009; López-Morales et al. 2010; Campo et al. 2011; Madhusudhan et al. 2011; Cowan et al. 2012; Crossfield et al. 2012; Copperwheat et al. 2013; Föhring et al. 2013; Sing et al. 2013; Swain et al. 2013; Stevenson et al. 2014a,b; Croll et al. 2015; Sing et al. 2016). This work presents the first optical eclipse measurement of WASP-12b.

The atmospheric composition of WASP-12b has been extensively studied (e.g., Madhusudhan et al. 2011; Crossfield et al. 2012; Swain et al. 2013; Stevenson et al. 2014b), with initial claims of a C/O ratio greater than unity. This was first challenged by Crossfield et al. (2012) and Cowan et al. (2012), who instead reported an isothermal photosphere for WASP-12b. The recent detection of water in the planet's atmosphere has now firmly refuted the carbon-rich hypothesis (Kreidberg et al. 2015). Sing et al. (2013) found that the best-fit model for WASP-12b transmission spectroscopy was Mie scattering by an aluminum-oxide (Al_2O_3) haze. Barstow et al. (2017) found that an optically thick Rayleigh scattering aerosol with a 0.01 mbar top pressure best described the transmission observations, but the model poorly described the steep increase in transit depth at optical wavelengths. Schwartz et al. (2017) used thermal phase variations and eclipse depths to determine a Bond albedo of $A_B = 0.2^{+0.1}_{-0.12}$ and a dayside effective temperature of $T_{\rm day} = 2864 \pm 15$ K.

2. OBSERVATIONS AND DATA REDUCTION

On 2016 October 19, a single eclipse of WASP-12b was observed with five *HST* orbits, using the STIS G430L grating (290–570 nm). The first *HST* orbit has significantly worse systematics than the four later orbits as a result of the repointing of the telescope, so these data were removed from the subsequent analysis. This left two *HST* orbits out of eclipse (one before and one after) when the planet and host star were both visible with the planet near full phase, as well as two *HST* orbits during eclipse when the planet was behind its host star, leaving only the star's light visible. These observations were granted as a part of programme GO-14797 (PI: Crossfield).

We used the same data collection method as previously used for similar observations (Sing et al. 2011, 2013, 2016; Evans et al. 2013). The subarray readout mode with a wide $52'' \times 2''$ slit was used to minimize time-varying slit losses; this produced 1024×128 pixel images. In previous *HST*/STIS observations, the first frame from each *HST* orbit had systematically lower counts, so a 1 s dummy exposure was obtained at the beginning of each orbit, which successfully mitigated this systematic effect. This dummy exposure was then followed by 10 science exposures lasting 279 s each (the maximum recommended duration to avoid excessive cosmic-ray hits). Our final, analyzed dataset thus contains 40 exposures collected over 331 minutes.

The raw STIS data were reduced (bias-, dark-, and flat-corrected) using the latest version of the `CALSTIS1` pipeline and the relevant up-to-date calibration frames.

1. 引言

对热木星的热测量表明，这些气态巨系外行星通常具有中等邦德反照率（$A_B \approx 0.4$，即入射能量反射至太空的比例；Schwartz等人2017年）。然而，先前许多针对热木星反射光的搜寻在宿主恒星释放大部分能量的光学波段发现极少甚至没有反射光（几何反照率$A_g < 0.1$；例如Rowe等人2008年；Kipping & Spiegel 2011年；Heng & Demory 2013年；Dai等人2017年）。目前尚不清楚是什么导致了热辐射约束与光学反射约束之间的明显矛盾。先前哈勃太空望远镜（HST）利用空间望远镜成像光谱仪（STIS）对HD 189733b进行的掩星观测显示，其反射率随波长变蓝而增加，这可能至少部分解释了两种技术之间的差异（Evans等人2013年）。

探测热木星大气层背向散射效率的直接方法是在行星处于全食期间（此时行星接近满相并运行至其宿主恒星后方）时，在光学波段（该波段热辐射可忽略）对其进行观测。这种方法所需的光度测量精度至少比观测同一行星的凌星现象高一个数量级，因为行星的亮度将低于其宿主恒星，而被遮挡的面积保持不变。

在不同轨道相位观测大气层，可进一步获取散射粒子的相关信息（如 Demory 等人 2013；Esteves 等人 2013；Heng & Demory 2013；Garcia Munoz & Isaak 2015；Shporer & Hu 2015；Oreshenko 等人 2016）。Parmentier 等人（2016）曾提出反射光相位曲线测量与一系列凝结云模型之间的关联，但其研究仅涵盖最高$T_{\rm eq} \sim 2200$ K 的温度范围：远低于 WASP-12b 的平衡温度（$T_{\rm eq} = 2580$ K；Collins 等人 2017）。

WASP-12b围绕一颗G0V型恒星运行，轨道周期为1.09天（Hebb等人，2009年）。尽管宿主星相当暗弱（V = 12），但WASP-12b的半长轴极近且半径巨大（$a = 0.0234$ au，$R_p = 1.90\ R_J$，$R_p = 0.19\ R_*$；Collins等人，2017年），使其成为进行详细研究的绝佳目标。对WASP-12b的凌星观测范围从0.3到4.5 μm，而掩星观测范围则从0.9到8.0 μm（例如，Hebb等人，2009年；López-Morales等人，2010年；Campo等人，2011年；Madhusudhan等人，2011年；Cowan等人，2012年；Crossfield等人，2012年；Copperwheat等人，2013年；Föhring等人，2013年；Sing等人，2013年；Swain等人，2013年；Stevenson等人，2014a,b；Croll等人，2015年；Sing等人，2016年）。本研究首次提供了WASP-12b的光学掩星测量结果。

WASP-12b的大气成分已被广泛研究（例如，Madhusudhan等人2011年；Crossfield等人2012年；Swain等人2013年；Stevenson等人2014b年），最初有观点认为其碳氧比大于1。这一观点首先受到Crossfield等人（2012年）和Cowan等人（2012年）的质疑，他们反而报告WASP-12b存在等温光球层。近期对该行星大气中水汽的探测现已明确推翻了富碳假说（Kreidberg等人2015年）。Sing等人（2013年）发现，WASP-12b透射光谱的最佳拟合模型是由氧化铝（Al_2O_3)雾霾引起的米氏散射。Barstow等人（2017年）发现，顶部压力为0.01毫巴的光学厚瑞利散射气溶胶能最好地描述透射观测结果，但该模型对光学波段透射深度陡增的现象解释不佳。Schwartz等人（2017年）利用热相位变化和食深度确定了邦德反照率为$A_B = 0.2^{+0.1}_{-0.12}$，昼侧有效温度为$T_{\rm day} = 2864 \pm 15$ K。

2. 观测与数据归算

2016年10月19日，利用STIS G430L光栅（290–570 nm）通过五次HST轨道观测到WASP-12b的一次凌星现象。由于望远镜重新定位的影响，第一次HST轨道的系统误差明显比后续四次轨道更大，因此这些数据在后续分析中被剔除。剔除后保留了两段HST非凌星轨道（凌星前后各一段），此时行星与宿主星均可见且行星接近满相；另有两段HST凌星轨道，期间行星位于宿主星后方，仅能观测到恒星光。这些观测数据属于GO-14797项目（首席研究员：Crossfield）的一部分。

我们采用了与先前类似观测相同的数据收集方法（Sing等人，2011、2013、2016；Evans等人，2013）。为尽量减少随时间变化的狭缝损失，我们使用了宽$52'' \times 2''$狭缝的子阵列读出模式，由此生成了1024× 128像素的图像。在以往的HST/STIS观测中，每个HST轨道的第一帧数据计数系统性偏低，因此我们在每个轨道开始时采集了1秒的虚拟曝光，成功缓解了这一系统效应。随后进行了10次科学曝光，每次持续279秒（这是为避免过多宇宙射线影响而推荐的最长持续时间）。最终，我们分析的数据集包含在331分钟内采集的40次曝光。

原始STIS数据使用最新版本的`CALSTIS1`流水线及相关的更新校准帧进行了处理（偏置、暗电流和平场校正）。

Cosmic-ray events were identified and removed following Nikolov et al. (2014), as were all pixels identified as "bad" by `CALSTIS`. Overall, $\sim$9% of the pixels in each 2D spectrum were affected by cosmic-rays with another $\sim$5% identified as "bad", resulting in a total of $\sim$14% interpolated pixels.

Next, the IRAF procedure `apall` was used to extract spectra from the calibrated .flt science files. We tested apertures between 9.0 and 17.0 pixels in intervals of 2 pixels and found that an 11.0 pixel aperture resulted in the lowest lightcurve residual scatter after fitting the white light data. However, the difference between apertures was minute ($\sim$1 ppm). We then used cross-correlation to correct for subpixel shifts along the dispersion axis. The x1d files from `CALSTIS` were then used to calibrate the wavelength axis. Finally, both "white light" and six spectral channel lightcurves were produced by integrating the appropriate flux from each bandpass.

WASP-12b's host star WASP-12A is also orbited by two M-dwarf companions bound in a binary system $1.06''$ away from WASP-12A (Bergfors et al. 2011; Sing et al. 2013; Bechter et al. 2014). For our observations, the spectrograph slit orientation was chosen to be perpendicular to the line connecting WASP-12A and WASP-12(B,C) to allow maximal separation in the spatial direction of the resulting FITS files. The spectrum of the stellar companions is visually distinguishable from WASP-12A in the raw spectra and does not fall within our small spatial-axis aperture.

3. LIGHTCURVE ANALYSIS

The top panel of Figure 1 shows the raw lightcurve binned across the entire STIS G430L bandpass ("white light"). There is a strong, repeated trend in flux, with exposures from each orbit appearing to follow a roughly polynomial trend. This systematic is well known and is believed to be the result of the thermal cycle of *HST* throughout its orbit as well as the movement of the spectral trace on the detector (e.g., Brown et al. 2001; Sing et al. 2011; Huitson et al. 2012; Evans et al. 2013).

These systematic trends are also observed during *HST*/STIS observations of planetary transits, and a standard approach to remove them is assuming polynomial variations as a function of auxiliary variables (e.g., Sing et al. 2011; Huitson et al. 2012). More recently, Gibson et al. (2011) used Gaussian processes (GPs) to model the *HST* systematics, as the choice of polynomial model can potentially bias the results. For this reason, we modelled the systematics with the GP library `george` (Foreman-Mackey 2015) and used the same method as Evans et al. (2013). A detailed discussion of modelling systematics with GPs can be found in Gibson et al. (2012a,b, 2013). We also attempted to fit the systematic variations with a polynomial model, which gave results consistent with our GP model.

3.1. *Gaussian Process Model*

The likelihood of a GP model is described as a multivariate normal distribution with

$$p(\boldsymbol{f}|\boldsymbol{X},\boldsymbol{\theta},\boldsymbol{\Omega},\boldsymbol{t}) = \mathcal{N}(\boldsymbol{E}(\boldsymbol{\Omega},\boldsymbol{t}),\boldsymbol{\Sigma}(\boldsymbol{X},\boldsymbol{\theta}))\,, \tag{1}$$

where $\boldsymbol{f}$ is the 40 measured fluxes, $\boldsymbol{E}$ is the eclipse function, and $\boldsymbol{\Sigma}$ is the *kernel* (covariance matrix). The time at the midpoint of each exposure is represented by $\boldsymbol{t}$. Further, $\boldsymbol{X} = [\boldsymbol{\phi},\boldsymbol{\psi}]^T$ is the matrix of covariates, where $\boldsymbol{\phi}$ is the orbital phase of *HST*, and $\boldsymbol{\psi}$ is the slope of the spectral trace on the detector (computed using IRAF's `apall` procedure). These two covariates were selected as they provided the lowest scatter in the residuals after calibration. We also tested the inclusion of two additional covariates: the y-intercept of the spectral trace on the detector and the measured shifts of the spectral trace along the dispersion axis. However, the inclusion of these additional covariates did not significantly impact our results or uncertainties, likely because the covariates themselves are significantly correlated with the other covariates.

Our eclipse parameters are given by $\boldsymbol{\Omega} = [\alpha,\delta,\beta]^T$, where α is the baseline flux consisting of light emitted from both the planet and star, δ is the fractional eclipse depth ($\delta = F_{\rm planet}/F_{\rm star}$), and β describes a constant rate of change in the baseline flux over time. Since we did not observe during eclipse ingress or egress, we used a boxcar function to describe the eclipse signal, with

$$\begin{aligned} E_i &= \alpha(1-\delta B_i)(1+\beta(t_i-t_0)) \\ B_i &= \begin{cases} 0 & i \in \text{Orbit 2 or 5} \\ 1 & i \in \text{Orbit 3 or 4}\,, \end{cases} \end{aligned} \tag{2}$$

where t_0 is the time of the first exposure.

Our GP parameters are given by $\boldsymbol{\theta} = [C, L_\phi, L_\psi, \sigma_w]^T$, where C^2 is the maximum covariance, L_ϕ and L_ψ are covariance lengthscales, and σ_w is the white noise level. We adopted the squared-exponential kernel:

$$\boldsymbol{\Sigma}_{nm} = C^2 \exp\left[-\sum_{i=0}^{1}\frac{(\boldsymbol{X}_{in}-\boldsymbol{X}_{im})^2}{L_i^2}\right] + \delta_{nm}\sigma_w^2\,, \tag{3}$$

where $L_i = [L_\phi, L_\psi]_i$ and δ_{nm} is the Kronecker delta function. This kernel can be simply understood as requiring that observations be strongly correlated if they have similar spectral trace slope and *HST* orbital phase,

宇宙射线事件被识别并移除，遵循Nikolov等人（2014）的方法，同时所有被`CALSTIS`标记为“坏”的像素也被移除。总体而言，每个二维光谱中~9%的像素受到宇宙射线影响，另有~5%被标记为“坏”像素，导致总共~14%的像素被插值处理。

接下来，使用IRAF程序`apall`从校准后的.flt科学文件中提取光谱。我们在9.0至17.0像素范围内以2像素为间隔测试了不同孔径，发现11.0像素孔径在拟合白光数据后能产生最低的光变曲线残差离散度。不过，不同孔径之间的差异极小（~1 ppm）。随后，我们采用互相关方法校正了沿色散方向的亚像素偏移。接着利用`CALSTIS`生成的x1d文件校准波长轴。最后，通过积分各波段的相应通量，生成了“白光”及六个光谱通道的光变曲线。

WASP-12b的宿主恒星WASP-12A还被两颗M型矮星伴星环绕，它们构成一个双星系统，距离WASP-12A约1.06″（Bergfors等人，2011；Sing等人，2013；Bechter等人，2014）。在我们的观测中，光谱仪狭缝方向被设置为垂直于连接WASP-12A与WASP-12(B,C)的连线，以便在最终FITS文件的空间方向上实现最大分离。恒星伴星的光谱在原始光谱中可与WASP-12A明显区分，且未落入我们设定的狭窄空间轴孔径范围内。

3. 光变曲线分析

图1的上图显示了在整个STIS G430L波段（"白光"）内合并的原始光变曲线。通量存在明显的重复趋势，每个轨道的曝光似乎都遵循大致多项式趋势。这种系统误差是众所周知的，被认为是{v*}在其整个轨道上的热循环以及光谱轨迹在探测器上移动的结果（例如，Brown等人，2001年；Sing等人，2011年；Huitson等人，2012年；Evans等人，2013年）。

这些系统性趋势在*HST*/STIS观测行星凌星期间同样被观察到，移除它们的标准方法是假设多项式变化作为辅助变量的函数（例如，Sing等人2011年；Huitson等人2012年）。最近，Gibson等人（2011年）使用高斯过程（GPs）来模拟*HST*系统误差，因为多项式模型的选择可能会使结果产生偏差。因此，我们采用GP库`george`(Foreman-Mackey 2015)对系统误差进行建模，并使用了与Evans等人（2013年）相同的方法。关于建模的详细讨论使用高斯过程进行系统误差分析的方法可以在Gibson等人（2012a,b, 2013）的研究中找到。我们还尝试用多项式模型拟合系统变化，其结果与我们的高斯过程模型一致。

3.1. *Gaussian Process Model*

GP模型的可能性被描述为一个多元正态分布，其参数为

$$p(\boldsymbol{f}|\boldsymbol{X},\boldsymbol{\theta},\boldsymbol{\Omega},\boldsymbol{t}) = \mathcal{N}(\boldsymbol{E}(\boldsymbol{\Omega},\boldsymbol{t}),\boldsymbol{\Sigma}(\boldsymbol{X},\boldsymbol{\theta}))\,, \tag{1}$$

其中 $\boldsymbol{f}$ 是40个实测流量，$\boldsymbol{E}$ 是凌食函数，$\boldsymbol{\Sigma}$ 是 *kernel*（协方差矩阵)。每次曝光中点时刻由 $\boldsymbol{t}$ 表示。此外，$\boldsymbol{X} = [\boldsymbol{\phi}, \boldsymbol{\psi}]^T$ 是协变量矩阵，其中 $\boldsymbol{\phi}$ 是 *HST* 的轨道相位，$\boldsymbol{\psi}$ 是光谱迹在探测器上的斜率（使用IRAF的 `apall` 程序计算）。选择这两个协变量是因为它们在校准后能使残差达到最小离散度。我们还测试了纳入另外两个协变量：光谱迹在探测器上的y轴截距以及光谱迹沿色散轴的实测偏移量。然而，纳入这些额外协变量并未显著影响我们的结果或不确定度，这可能是因为这些协变量本身与其他协变量存在显著相关性。

我们的掩星参数由 $\boldsymbol{\Omega} = [\alpha,\delta,\beta]^T$ 给出，其中 α 是包含行星和恒星所发光量的基线流量，δ 是掩星深度分数（$\delta = F_{\rm planet}/F_{\rm star}$），而 β 描述了基线流量随时间变化的恒定速率。由于我们在掩星进入或退出阶段未进行观测，因此采用一个矩形函数来描述掩星信号，其中

$$\begin{aligned} E_i &= \alpha(1-\delta B_i)(1+\beta(t_i - t_0)) \\ B_i &= \begin{cases} 0 & i \in \text{Orbit 2 or 5} \\ 1 & i \in \text{Orbit 3 or 4}\,, \end{cases} \end{aligned} \tag{2}$$

其中t_0为首次曝光的时间。

我们的高斯过程参数由 $\boldsymbol{\theta} = [C, L_\phi, L_\psi, \sigma_w]^T$ 给出，其中 C^2 是最大协方差，L_ϕ 和 L_ψ 是协方差长度尺度，σ_w 是白噪声水平。我们采用了平方指数核：

$$\boldsymbol{\Sigma}_{nm} = C^2 \exp\left[-\sum_{i=0}^{1} \frac{(\boldsymbol{X}_{in} - \boldsymbol{X}_{im})^2}{L_i^2}\right] + \delta_{nm}\sigma_w^2\,, \tag{3}$$

其中$L_i = [L_\phi, L_\psi]_i$且δ_{nm}为克罗内克δ函数。该核函数可简单理解为：若观测数据具有相似的光谱迹线斜率与*HST*轨道相位，则要求它们之间必须存在强相关性。

while observations further from each other in covariate space are more weakly correlated. This then describes a smoothly varying function of the covariates, with the addition of white noise.

The final model is then given by

$$f^* = \boldsymbol{\mu}(\boldsymbol{\phi}, \boldsymbol{\psi}) + \boldsymbol{E}(\boldsymbol{\Psi}), \tag{4}$$

where $\boldsymbol{\mu}(\boldsymbol{\phi}, \boldsymbol{\psi})$ is the GP model mean.

The Markov Chain Monte Carlo (MCMC) ensemble sampler software `emcee` (Foreman-Mackey et al. 2013) was used to explore the seven parameters, determining the most likely parameter values and their uncertainties. For computational reasons, the variables used in this MCMC were $\{\delta,\ \ln(\alpha),\ \beta,\ \ln(L_\phi),\ \ln(L_\psi),\ \ln(\sigma_w^2),$ and $\ln(C^2)\}$. Using logarithms removes the need to use a prior to obtain strictly positive values. While the eclipse depth, δ, should be strictly positive, we allowed for negative values to ensure an unbiased estimate. A uniform prior was used so $\ln(L_\phi) < 0$ and $\ln(L_\psi) < 0$, which has the effect of ensuring that these lengthscales are within a few orders of magnitude of the variations in the covariates. For un-normalized white light data, the best-fit values from a 10^6 step MCMC chain were $\{\delta = (-5.3 \pm 7.4) \times 10^{-5}$, $\alpha = (4.198^{+0.012}_{-0.008}) \times 10^7$, $\beta = 0.0056 \pm 0.0015$, $L_\phi = 0.05^{+0.19}_{-0.03}$, $L_\psi = 0.03^{+0.27}_{-0.03}$, $\sigma_w^2 = (5.2^{+1.7}_{-1.3}) \times 10^7$, and $C^2 = (8^{+14}_{-5}) \times 10^7\}$.

The top panel of Figure 1 shows the median model and uncertainty from a 10^6 step MCMC chain overplotted on the raw white light flux measurements. The bottom panel of Figure 1 shows the lightcurve produced by dividing the raw spectra by the median model (excluding the change in flux during eclipse), with the median eclipse model overplotted. The clear linear trend in the calibrated flux of *HST* orbit #5 (bottom panel of Figure 1) shows that there is still substantial correlated noise in the data that could not be described by any of the four considered covariates.

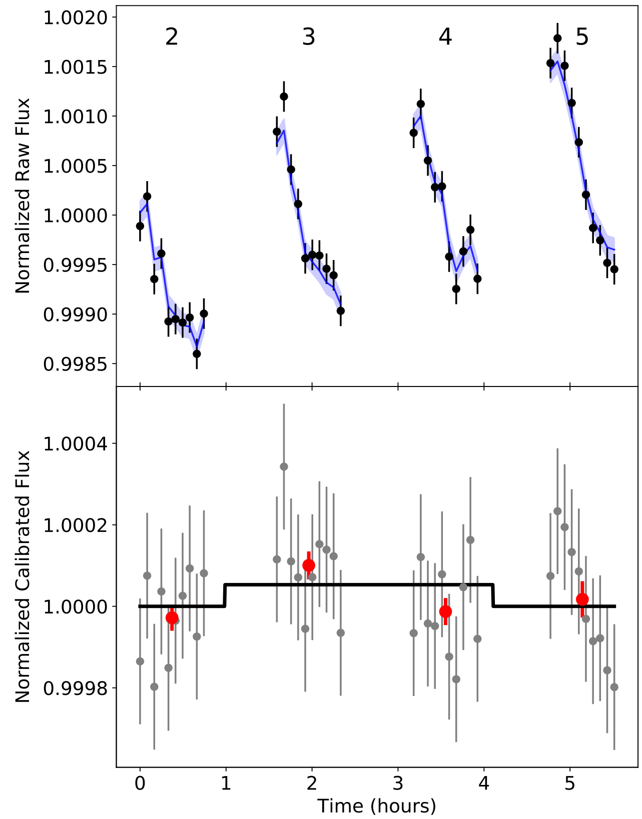

Figure 1. *Top:* raw flux with the entire spectral range of *HST*/STIS binned into a single white lightcurve. The median systematic model and 1σ model uncertainty are shown with a blue line and blue shaded region, respectively. Each individual *HST* orbit is labelled. *Bottom:* the white light data after calibration using a Gaussian Process are shown in grey. Also shown in red are the binned fluxes for each *HST* orbit, although these were not used during fitting. Overplotted is the best-fit eclipse signal that corresponds to a wavelength-averaged geometric albedo of $A_g = -0.035$ ($A_g < 0.064$ at 97.5% confidence). All plotted error bars in both panels only capture uncorrelated, white noise.

4. RESULTS

The STIS G430L spectra were binned into six spectral channels to allow moderate wavelength resolution while keeping uncertainties on each channel sufficiently small to be able to test atmospheric models. Each spectral channel was modelled independently using the GP method described above. Lightcurves after GP calibration are shown for each spectral channel in Figure 2, and the relevant results are tabulated in Table 1. Eclipse depths were found using the median value from a 10^6 step MCMC chain, while the 84 and 97.5 percentiles were used to determine upper limits. The larger uncertainties in eclipse depth at shorter wavelengths are due to lower stellar flux and detector sensitivity.

Because WASP-12b is so strongly irradiated, the peak of its thermal emission is expected to be at $\sim$1 μm for a $\sim$2800 K dayside temperature (Schwartz et al. 2017). For this reason, we calculated the predicted eclipse depths due to thermal radiation from WASP-12b, assuming a $T = 3000$ K blackbody for WASP-12b (hotter than inferred from infrared observations due to the greater depth of the optical photosphere; Cowan & Agol 2011) and a standard G0V spectrum from Pickles (1998) for WASP-12. These depths ($\delta_{\rm thermal}$) are summarized in Table 1 and are all within our 97.5% confidence interval upper limits.

而协变量空间中彼此距离较远的观测值相关性较弱。这便描述了一个随协变量平滑变化的函数，并附加了白噪声。

最终模型由下式给出

$$f^* = \boldsymbol{\mu}(\boldsymbol{\phi}, \boldsymbol{\psi}) + \boldsymbol{E}(\boldsymbol{\Psi}), \tag{4}$$

其中$\boldsymbol{\mu}(\boldsymbol{\phi}, \boldsymbol{\psi})$是GP模型的均值。

采用马尔可夫链蒙特卡洛（MCMC）集成采样软件 `emcee` (Foreman-Mackey 等人 2013) 对七个参数进行探索，以确定最可能的参数值及其不确定性。出于计算考虑，本次 MCMC 中使用的变量为 $\{\delta$、$\ln(\alpha)$、β、$\ln(L_\phi)$、$\ln(L_\psi)$、$\ln(\sigma_w^2)$ 和 $\ln(C^2)\}$。使用对数可避免引入先验分布来强制获取严格正值。虽然食深 δ 应为严格正值，但为获得无偏估计我们允许其取负值。采用均匀先验分布使得 $\ln(L_\phi) < 0$ 且 $\ln(L_\psi) < 0$，这能确保这些长度尺度与协变量变化量级相差在数个数量级内。对于未归一化的白光数据，经过 10^6 步 MCMC 链得到的最佳拟合值为 $\{\delta = (-5.3 \pm 7.4) \times 10^{-5}$、$\alpha = (4.198^{+0.012}_{-0.008}) \times 10^7$、$\beta = 0.0056 \pm 0.0015$、$L_\phi = 0.05^{+0.19}_{-0.03}$、$L_\psi = 0.03^{+0.27}_{-0.03}$、$\sigma_w^2 = (5.2^{+1.7}_{-1.3}) \times 10^7$ 以及 $C^2 = (8^{+14}_{-5}) \times 10^7\}$。

图1的上图展示了经过10^6步MCMC链计算得到的中值模型及其不确定性，该模型叠加在原始白光通量测量数据上。下图则显示了将原始光谱除以中值模型（排除食相期间通量变化）后生成的光变曲线，并叠加了中值食相模型。HST第5轨道的校准通量中明显的线性趋势（图1下图）表明，数据中仍存在显著的相关噪声，且这些噪声无法通过所考虑的四项协变量中的任何一项来描述。

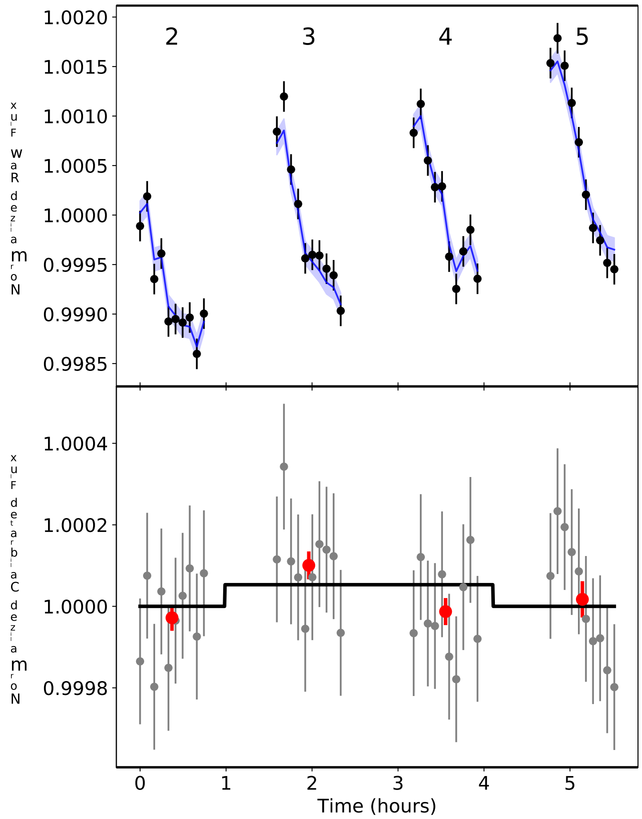

Figure 1. *Top:* 原始通量，将HST/STIS的整个光谱范围合并为单一的白光曲线。中值系统模型和1σ模型不确定性分别用蓝线和蓝色阴影区域表示。每个独立的HST轨道均已标注。*Bottom:*使用高斯过程校准后的白光数据以灰色显示。此外，红色显示的是每个HST轨道的合并通量，尽管这些数据在拟合过程中未被使用。图中叠加了最佳拟合的掩食信号，对应波长平均几何反照率为$A_g = -0.035$（97.5%置信度下为$A_g < 0.064$）。两个面板中所有绘制的误差条仅包含不相关的白噪声。

4. 结果

STIS G430L光谱被分成了六个光谱通道，以在保持每个通道不确定性足够小的同时，获得适中的波长分辨率，从而能够测试大气模型。每个光谱通道都使用上述GP方法独立建模。图2展示了每个光谱通道经过GP校准后的光变曲线，相关结果列于表1。食深是通过10^6步MCMC链的中值确定的，而84和97.5百分位数则用于确定上限。较短波长处食深的不确定性较大，这是由于恒星通量较低和探测器灵敏度不足所致。

由于WASP-12b受到强烈的辐射，根据其昼侧温度$\sim$2800 K（Schwartz等人，2017年），其热辐射峰值预计在$\sim$1 μm处。因此，我们计算了WASP-12b热辐射导致的预测掩食深度，其中假设WASP-12b为T =3000 K的黑体（由于光学光球层深度更大，此温度高于红外观测的推断值；Cowan & Agol 2011），并采用Pickles（1998）的标准G0V光谱作为WASP-12的模型。这些深度值（$\delta_{\rm thermal}$）总结于表1中，且均在我们97.5%置信区间的上限范围内。

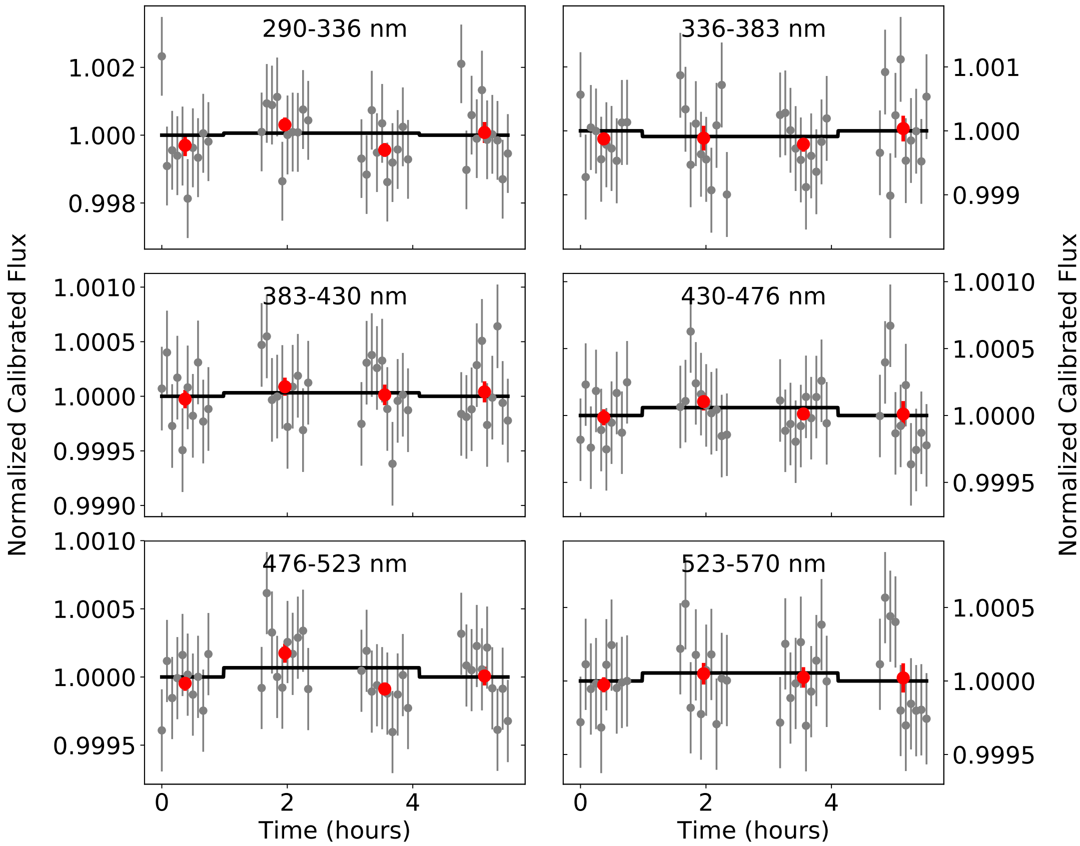

Figure 2. Lightcurves for each spectral channel after calibration using a Gaussian process are shown in grey, with the best-fit eclipse signal overplotted. Also shown in red are the binned fluxes for each *HST* orbit, although these were not used during fitting. All plotted error bars only capture uncorrelated, white noise; where no error bar is visible, it is smaller than the point size used.

If interpreted as solely due to reflected light, eclipse depths can be converted to geometric albedo using

$$A_g = \delta \left(\frac{R_p}{a} \right)^{-2}, \quad (5)$$

where $R_p = 1.90\ R_J$ is the radius of the planet, and $a = 0.0234$ AU is its orbital semi-major axis (Collins et al. 2017). Applying Equation (5) to the best-fit eclipse depths and their corresponding upper limits gives constraints on the geometric albedo across the STIS G430L wavelength range (summarized in Table 1)

Our reported uncertainties on eclipse depths are $\sim$40% higher than the photon limit. The increased scatter in our data may be the result of incomplete modelling of the systematic noise. Alternatively, our uncertainties may be limited by intrinsic stellar variability. Given the slow rotation period of WASP-12 compared to the observing window ($P_{rot} \gtrsim 23$ days given $v \sin i < 2.2$ km/s; Hebb et al. 2009), variability due to stellar rotation (e.g. starspots passing in and out of view) should not significantly affect our observations. However, our Sun's total irradiance (spatially and spectrally integrated) is known to vary at the 10^{-4} level on timescales of minutes to hours as a result of solar convection and oscillations (Kopp 2016). Given the G0V spectral class of WASP-12, similar variations may also be present and may explain the greater than Poisson limit uncertainties as well as the residual correlated noise in the calibrated time-series spectra.

5. DISCUSSION AND CONCLUSIONS

We use the `NEMESIS` spectral retrieval tool (Irwin et al. 2008; Barstow et al. 2014) to produce predicted model spectra given two previously proposed models for WASP-12b: an Al_2O_3 haze and a cloud-free atmosphere. `NEMESIS` is not a radiative equilibrium code; rather, it takes an atmospheric model and calculates incident and emergent flux and will not take into account heating from incoming stellar radiation. The limits from our *HST*/STIS eclipse observations firmly reject both models; we find χ^2 per datum (χ^2/N_{obs}, $N_{obs} = 6$) of 41 and 10 for the Al_2O_3 haze and cloud-free models.

Given its exceedingly high equilibrium temperature ($T_{\rm eq} = 2580$ K; Collins et al. 2017), WASP-12b would

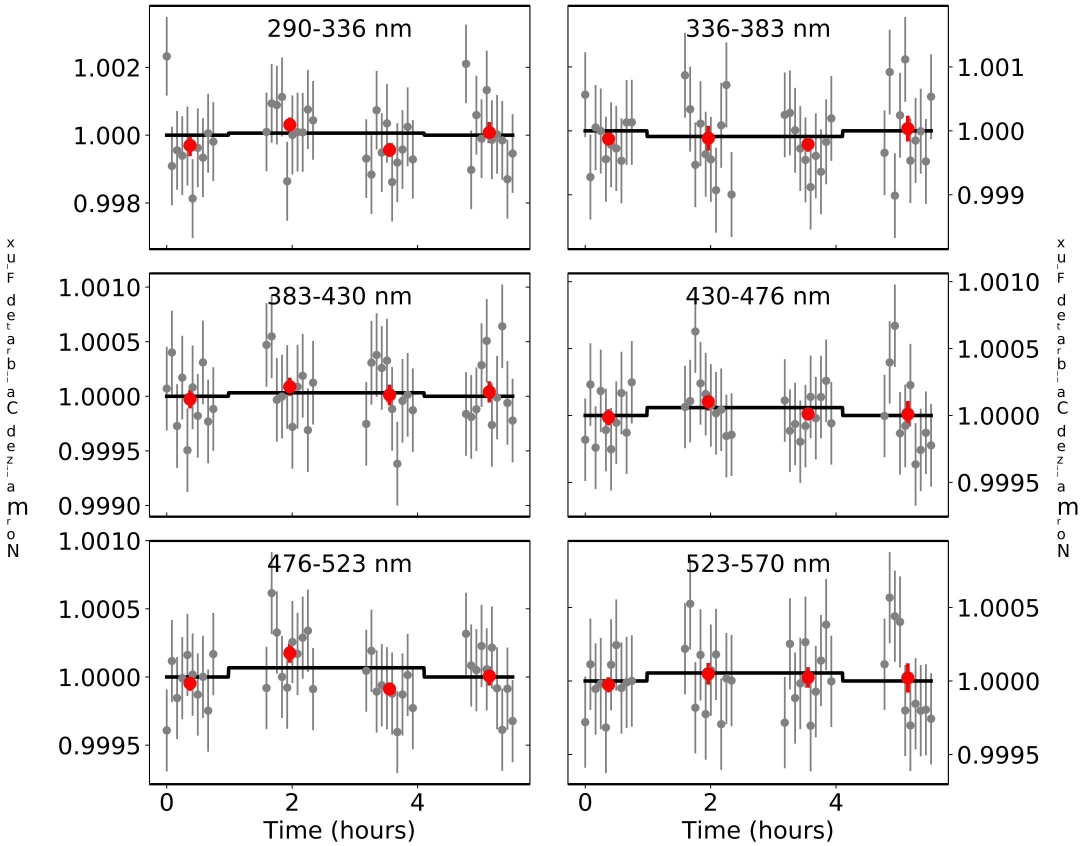

Figure 2. 使用高斯过程校准后，每个光谱通道的光变曲线以灰色显示，并叠加了最佳拟合的掩食信号。此外，每个HST轨道的分箱通量以红色显示，尽管这些数据在拟合过程中并未使用。所有绘制的误差条仅包含不相关的白噪声；若未显示误差条，则表示其小于所使用的点尺寸。

如果仅解释为反射光所致，则可以利用{v*}将食深转换为几何反照率。

$$A_g = \delta\left(\frac{R_p}{a}\right)^{-2}, \tag{5}$$

其中R_p =1.90R_J为行星半径，a =0.0234 AU为其轨道半长轴（Collins等人，2017年）。将方程（5）应用于最佳拟合的食深及其对应的上限，可得出STIS G430L波长范围内几何反照率的约束条件（总结于表1）。

我们报告的掩星深度不确定性比光子极限高出~40%。我们数据中增加的离散性可能是系统噪声建模不完整的结果。或者，我们的不确定性可能受到恒星本身固有变化的限制。考虑到WASP-12的自转周期相对于观测窗口较慢（根据v sin i < 2.2 km/s给出的$P_{rot} \gtrsim$ 23天；Hebb等人，2009年），由恒星自转引起的变异性（例如星斑进出视野）不应显著影响我们的观测。然而，我们太阳的总辐照度（空间和光谱积分）是已知由于太阳对流和振荡（Kopp 2016），在几分钟到几小时的时间尺度上会出现10^{-4}级别的变化。考虑到WASP-12的G0V光谱型，类似的变化可能同样存在，这或许能解释超出泊松极限的不确定性，以及校准时间序列光谱中残留的相关噪声。

5. 讨论与结论

我们使用NEMESIS光谱检索工具（Irwin等人，2008；Barstow等人，2014），基于先前提出的两种WASP-12b模型——Al_2O_3雾霾模型与无云大气模型——生成预测光谱。NEMESIS并非辐射平衡代码；其工作原理是接收大气模型并计算入射与出射通量，且不会考虑来自恒星辐射的加热效应。我们的HST/STIS掩星观测数据明确排除了这两种模型：针对Al_2O_3雾霾模型与无云模型，我们得出每个数据点的χ^2值（χ^2/N_{obs}，$N_{obs}=6$）分别为41和10。

鉴于其极高的平衡温度（T_{eq} = 2580 K；Collins 等人，2017 年），WASP-12b 将

Table 1. Eclipse Depths and Geometric Albedos

Wavelengths	Eclipse Depth, δ (ppm)		δ_{thermal}	Geometric Albedo, A_g	
(nm)	Best fit	97.5% Upper Limit	(ppm)	Best Fit	97.5% Upper Limit
290 – 570	-53 ± 74	96	56	-0.035 ± 0.050	0.064
290 – 336	-60 ± 540	1020	10	-0.04 ± 0.36	0.68
336 – 383	90 ± 290	670	20	0.06 ± 0.20	0.45
383 – 430	-30 ± 180	330	40	-0.02 ± 0.12	0.22
430 – 476	-60 ± 130	210	60	-0.039 ± 0.089	0.14
476 – 523	-70 ± 130	190	100	-0.045 ± 0.087	0.13
523 – 570	-50 ± 150	240	160	-0.036 ± 0.098	0.16

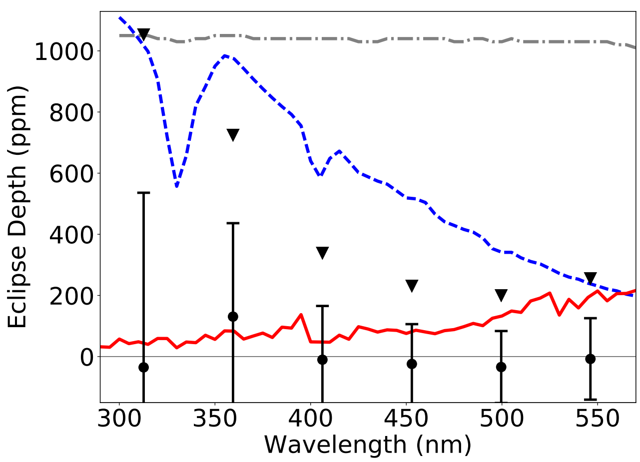

Figure 3. Best-fit eclipse depths and 1σ uncertainties are shown with black points and error bars, with black triangles denoting 97.5% confidence upper limits. Previously proposed models for WASP-12b made with `NEMESIS` are shown with a grey, dashed-dotted line (aluminum-oxide haze) and a blue, dashed line (cloud-free). The *HST*/STIS data firmly reject both models and are instead consistent with the thermally dominated `PHOENIX` model shown with a red solid line.

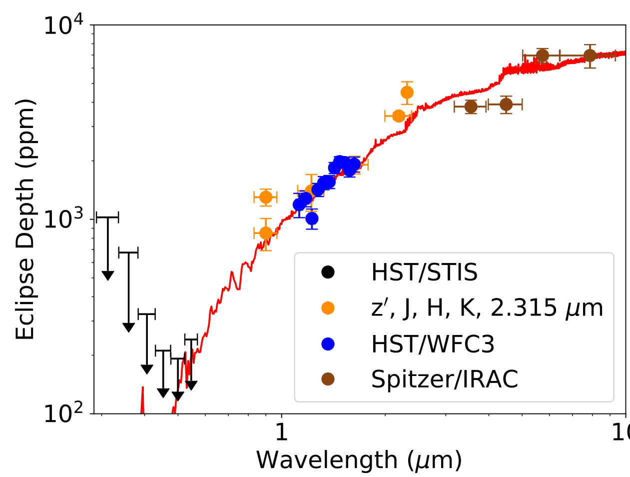

Figure 4. Our *HST*/STIS 97.5% confidence interval upper limits on the eclipse depth for each of the six considered spectral channels are shown with black arrows. All eclipse depths aside from *HST*/STIS are taken from Stevenson et al. (2014b, and references therein). The red line is the same as in Figure 3.

technically lie within Sudarsky et al.'s (2000) Class V ($T_{\rm eff} > 1500$ K) but is far hotter than any planet they considered. On the planet's dayside, WASP-12b is far too hot for condensates to form (Wakeford et al. 2017). However, temperatures near the planet's day–night terminator, and across the planet's nightside, may be cool enough to allow for the formation of condensates that could affect transmission spectroscopy without significantly affecting dayside eclipse spectroscopy.

Also, it is expected that Na I absorption (which is important at lower temperatures) will not contribute much to the low albedo of WASP-12b as most of the sodium will be ionized on the hot dayside. Instead, it is expected that the atmosphere will be dominated by Rayleigh scattering from atomic hydrogen and helium, with a small contribution from electron scattering. The red line in Figures 3 and 4 shows the predicted eclipse depth (binned to a resolution of 1 point per 5 nm) from Crossfield et al. (2012) made with the `PHOENIX` atmosphere code adapted for hot Jupiters as described in Barman et al. (2001, 2005). In this model, reflected light makes up $\lesssim 10\%$ of the eclipse depth ($A_g \lesssim 0.002$) at the shortest wavelengths and $\ll 1\%$ of the eclipse depth at infrared wavelengths; the remainder of the eclipse depth is due to thermal emission. This model gives a χ^2 per datum of 0.9 for our *HST*/STIS data ($N_{obs} = 6$), but a

Table 1. 日食深度与几何反照率

Wavelengths	Eclipse Depth, δ (ppm)		$\delta_{\mathrm{thermal}}$	Geometric Albedo, A_g	
(nm)	Best fit	97.5% Upper Limit	(ppm)	Best Fit	97.5% Upper Limit
290 – 570	-53 ± 74	96	56	-0.035 ± 0.050	0.064
290 – 336	-60 ± 540	1020	10	-0.04 ± 0.36	0.68
336 – 383	90 ± 290	670	20	0.06 ± 0.20	0.45
383 – 430	-30 ± 180	330	40	-0.02 ± 0.12	0.22
430 – 476	-60 ± 130	210	60	-0.039 ± 0.089	0.14
476 – 523	-70 ± 130	190	100	-0.045 ± 0.087	0.13
523 – 570	-50 ± 150	240	160	-0.036 ± 0.098	0.16

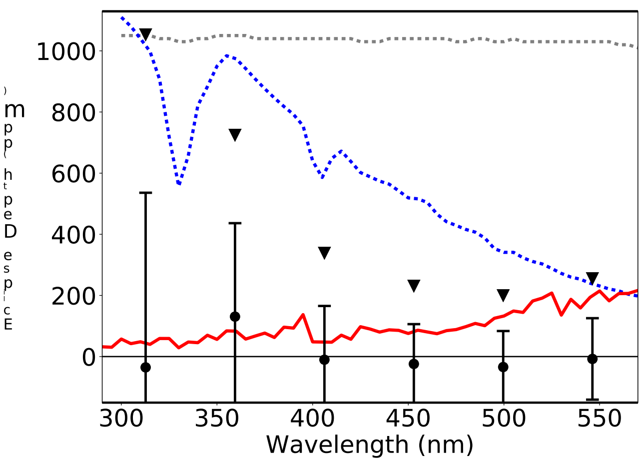

Figure 3. 最佳拟合的掩食深度和1σ不确定性以黑色点和误差条显示，黑色三角形表示97.5%置信度的上限。先前使用NEMESIS为WASP-12b提出的模型以灰色点划线（氧化铝雾霾）和蓝色虚线（无云）显示。*HST*/STIS数据明确否定了这两个模型，反而与以红色实线显示的热主导PHOENIX模型一致。

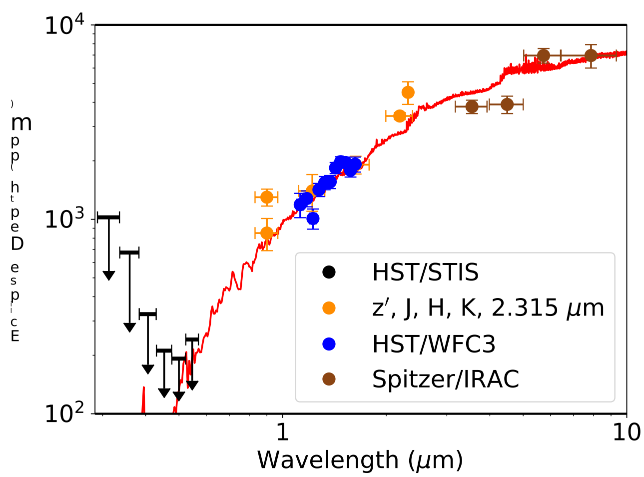

Figure 4. 我们的*HST*/STIS 97.5%置信区间上界（对应六个考虑的光谱通道各自的掩食深度）以黑色箭头标示。除*HST*/STIS外，所有掩食深度数据均取自Stevenson等人（2014b及其参考文献）。红线与图3中所示一致。

从技术上讲，它属于苏达斯基等人（2000年）提出的第五类行星（{v*} 1500 K），但比他们所考虑的任何行星都要热得多。在WASP-12b的昼侧，温度过高以至于无法形成凝结物（Wakeford等人，2017年）。然而，在行星昼夜交界线附近以及整个夜侧，温度可能足够低，允许形成凝结物，这些凝结物可能影响透射光谱，而不会显著影响昼侧掩星光谱。

此外，预计Na I吸收（在较低温度下较为重要）不会对WASP-12b的低反照率产生太大贡献，因为大部分钠在炽热的昼侧会被电离。相反，它

预计大气层将主要由原子氢和氦的瑞利散射主导，电子散射的贡献较小。图3和图4中的红线显示了Crossfield等人（2012年）预测的掩食深度（按每5纳米1个点的分辨率进行分箱），该预测使用专为热木星调整的PHOENIX大气代码（如Barman等人（2001年，2005年）所述）得出。在此模型中，反射光在最短波长处占掩食深度的$\lesssim$10%（$A_g \lesssim$0.002），在红外波长处占掩食深度的$\ll$1%；其余掩食深度由热辐射贡献。该模型为我们的*HST*/STIS数据（N_{obs} =6）给出的每个数据点的χ^2为0.9，但

worse χ^2 per datum of 3 for all of the data plotted on Figure 4 ($N_{obs} = 21$).

There are significant differences between the PHOENIX model and the cloud-free model produced by NEMISIS, including but not limited to the inclusion of atomic hydrogen opacities (lines and bound-free opacities), as well as the typical opacities more commonly associated with cool stellar photospheres. Also, the PHOENIX model results from a self-consistent calculation of the thermal structure, chemistry, line-by-line opacities (as well as scattering), and irradiation, thereby accounting for important changes that occur in the hot upper layers of WASP-12b (for example, the transition from H_2 to H at low pressures and the thermal ionization of Na and K).

Our observations cover the blackbody peak of WASP-12 (∼450 nm) and show that little of the incident radiation at these wavelengths is reflected by the planet. Geometric albedo is related to spherical albedo through a phase integral q such that $A_s = qA_g$, and Bond albedo is equal to the flux-weighted, wavelength-averaged spherical albedo. If we assume diffuse scattering ($q = 1.5$), our "white light" 97.5% confidence upper limit on the geometric albedo across the STIS bandpass ($A_g < 0.064$) suggests < 10% of the energy received at these wavelengths is reflected. However, since the wavelengths observed cover only 36% of the incident stellar energy, the Bond albedo is not well constrained by these measurements and is consistent with Schwartz et al.'s (2017) measurement of $A_B = 0.2^{+0.1}_{-0.12}$.

Our results are in stark contrast with those for the much cooler HD 189733b, the only other hot Jupiter with spectrally resolved reflected light observations (Evans et al. 2013); those data showed an increase in albedo with decreasing wavelength. The fact that the first two exoplanets with optical albedo spectra exhibit significant differences demonstrates the importance of spectrally resolved reflected light observations and highlights the great diversity among hot Jupiters.

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Facilities: HST(STIS)

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图4中绘制的所有数据的每个数据点3的χ^2更差（$N_{obs} =$ 21）。

PHOENIX模型与NEMISIS生成的云层模型存在显著差异，这些差异包括但不限于原子氢不透明度（谱线和束缚-自由不透明度）的纳入，以及更常见于冷恒星光球的典型不透明度。此外，PHOENIX模型源自对热结构、化学组成、逐线不透明度（以及散射）和辐射照射的自洽计算，从而解释了WASP-12b高温上层发生的重要变化（例如，低压下从H_2到H的转变以及Na和K的热电离）。

我们的观测覆盖了WASP-12的黑体辐射峰值（∼450纳米），并表明在这些波长上，入射辐射很少被行星反射。几何反照率通过相位积分q与球面反照率相关联，即$A_s = qA_g$，而邦德反照率等于通量加权、波长平均的球面反照率。如果我们假设漫散射（$q = 1.5$），我们在STIS波段（$A_g < 0.064$）的“白光”97.5%置信上限几何反照率表明，在这些波长接收的能量中只有< 10%被反射。然而，由于观测波长仅覆盖入射恒星能量的36%，邦德反照率受这些测量的约束较弱，且与Schwartz等人（2017）测得的$A_B = 0.2^{+0.1}_{-0.12}$一致。

我们的结果与温度低得多的HD 189733b形成鲜明对比，后者是唯一另一个拥有光谱分辨反射光观测数据的热木星。

(Evans et al. 2013)；这些数据显示反照率随波长减小而增加。前两颗拥有光学反照率谱的系外行星表现出显著差异，这一事实证明了光谱分辨反射光观测的重要性，并凸显了热木星之间的巨大多样性。

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Facilities: 哈勃太空望远镜（空间望远镜成像光谱仪）

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